Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **data engineering** pipelines on **Databricks**, Spark, AWS Glue, and SQL warehouses, shipping AI **education** platforms for instructors and learners, and **presenting** technical work to clients and conference audiences.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Database Systems, Advanced NLP, Web Technologies, Analysis of Algorithms

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Operating Systems, Object-Oriented Programming

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*
- Built **ETL**-style integration pipelines across PostgreSQL, SQL Server, Supabase, and cloud storage with incremental loads, data validation, and reconciliation, running 5K+ monthly jobs at 99%+ production reliability.
- Designed multi-tenant schemas aligned with **data warehouse** principles for an AI education platform serving 500+ users, cutting dashboard latency from 1.2s to 350ms through indexing and query tuning.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*
- Built an AI **assessment** platform for 150+ instructors that generates rubrics from reference answers and grades text and handwritten submissions with GPT-4o, Claude, and Qwen-VL at 90%+ consistency.
- Designed **instructor-facing** workflows for question upload, grading history, and feedback delivery on React and Flask, handling 10K+ daily requests, processing 1K+ files per batch, and cutting feedback turnaround 60%.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*
- **Led a team** redesigning ETL workflows on AWS Glue and S3 that normalized raw JSON, CSV, and Parquet data, boosting pipeline efficiency 68% and cutting storage costs 15%.
- Collaborated with Microsoft and GitHub **stakeholders** in Agile sprint meetings while automating 20+ nightly unit and integration test suites with emailed coverage reports across three operating systems.

Pandit Deendayal Energy University **January 2023 – May 2024**
Research Fellowship *India*
- Designed a **quantum-resistant blockchain** pairing lattice-based FALCON signatures with node-monitored delegated proof-of-stake consensus and Borda Count witness selection, cutting centralized-control and quantum-risk exposure 25% in benchmark simulations.
- **Presented** benchmark results at IISc Bengaluru and co-authored an IEEE paper, comparing the protocol with an ECDSA baseline at 957.8 versus 968.3 TPS across 100–1,000 transaction workloads.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*
- Built hourly **Databricks** and Spark pipelines ingesting 50K+ Firebase booking and payment events in micro-batches, aggregating peak hours, service demand, daily active users, and revenue for client reporting.
- **Presented** the end-to-end architecture and 12+ integration points to the client for alignment, then delivered Flask APIs with Firebase real-time sync sustaining sub-200ms latency across 50K+ records.

Skills

Data Engineering: Databricks, Apache Spark, Snowflake, AWS Glue, ETL/ELT, Data Warehousing, Data Modeling, Incremental Loads, Data Validation, Reconciliation
Databases & SQL: SQL, PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB, Query Optimization, Indexing
Languages & Tools: Python, Java, TypeScript, JavaScript, Git, Docker, AWS (S3, RDS), Azure, CI/CD, Jenkins
AI & EdTech: LLM Applications, Prompt Chaining, Automated Grading, Rubric Generation, RAG, Embeddings, Claude Code
Teaching & Communication: Technical Presentations, Technical Writing, Team Leadership, Mentoring, Requirements Gathering, Stakeholder Communication

Projects

Snowflake and Databricks Data Engineering Labs | *Snowflake, Databricks, Spark SQL, Python, SQL, AWS S3*
- Designed a persona-based **data engineering curriculum** with hands-on labs loading S3 data into Snowflake and Databricks Delta tables, auto-graded Python and SQL assessments, and capstone projects for analysts and engineers.

CPU-Scheduling Simulator Live | *React.js, JavaScript, EJ-2, C++*
- Built an **educational** CPU-scheduling simulator with interactive process inputs, EJ-2 Gantt charts, and live waiting and turnaround metrics for six algorithms, helping learners visualize scheduling across 1K+ scenarios.

Chat2SQL Code | *Java, TypeScript, Snowflake, PostgreSQL, Docker*
- Extended an AI-assisted SQL workspace on Java and TypeScript supporting 10+ databases including **Snowflake** and PostgreSQL, generating schema-aware queries at 82% accuracy and AI-generated dashboards for faster data exploration.